RELEASE COMPLIANCE DOCUMENT

RIVERSIDE OS

Riverside OS v0.80.8 Printable Deployment Guide

Authoritative reference manual for store hardware configuration, software package updates, schema migrations, and backup verification.

TARGET VERSION	v0.80.8
DOCUMENT CLASSIFICATION	Operational Standard Work
AUTHORIZED STATIONS	Server PC / Register #1 / Register #2 (iPad) / Counterpoint PC
GENERATED ON	May 27, 2026

Table of Contents

Use this guide for a live-store install or recovery pass. It is written in the order the work should happen:

1. Confirm deployment model and store network facts.
2. Verify release files and download required assets.
3. Install and verify the Backoffice / Server PC.
4. Configure all external integrations (QBO, Helcim, Shippo, Podium, Meilisearch, Counterpoint, ROSIE).
5. Deploy and configure Register #1 (Windows POS App).
6. Verify peripherals (receipt printer, cash drawer, scanner) on Register #1.
7. Deploy Register #2 PWA on iPad.
8. Set up additional Back Office workstation interfaces.
9. Perform live production smoke test and test sale.
10. Confirm backups, database safety, and rollback controls.

Print this guide and check off each section during deployment.

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1. Deployment Model

Store Roles

Station	Device	What it does
Backoffice / Server PC	Windows PC	Runs PostgreSQL, Riverside Server API, web/PWA files, migrations, firewall rule, backups, and the startup task. Keep this PC on.
Register #1	Windows PC	Main cashier station. Owns the receipt printer, cash drawer, and till group close.
Register #2	iPad PWA	Satellite register lane. Uses the server over the store network and attaches to the Register #1 till group.
Other Back Office PCs	Browser/PWA or desktop app	Back-office work against the same server. Do not install the server role here.
Counterpoint PC	Existing Windows/SQL host	Runs the Counterpoint bridge only when importing or syncing legacy Counterpoint data.

API Host Rule

Use the right API host before troubleshooting anything else.

Device	API host value
Backoffice / Server PC	`http://127.0.0.1:3000`
Register #1 Windows PC	`http://SERVER-IP:3000`, for example `http://10.64.70.196:3000`
Register #2 iPad PWA	`http://SERVER-IP:3000`, for example `http://10.64.70.196:3000`
Other Back Office PCs	`http://SERVER-IP:3000` or trusted HTTPS/Tailscale URL
Counterpoint bridge `.env`	`ROS_BASE_URL=http://SERVER-IP:3000`

Do not use `127.0.0.1` on Register #1, iPad, or the Counterpoint bridge unless Riverside Server is running on that same device.

Register Cash Rule

- Register #1 is the physical cash drawer lane.
- Register #2 and other satellite lanes may enter sales, but physical cash still belongs to Register #1.
- Register #1 performs the till group close / Z-close.

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2. Release Files And Records

Use only the official GitHub release files for the main install:

```
https://github.com/cpg716/riverside-os/releases/tag/v0.80.8
```

Required release assets:

- [ ] `RiversideOS-v0.80.8-Windows-Deployment.zip`
- [ ] `Riverside.POS_0.80.8_x64_en-US.msi`
- [ ] `Riverside.POS_0.80.8_x64_en-US.msi.sig`
- [ ] `latest.json`
- [ ] `riverside-updater-build-manifest.json`
- [ ] `RiversideOS-Deployment-Manager_0.80.8_universal.dmg` (macOS remote management)

Expected release identity:

```
Version: 0.80.8
Deployment package: RiversideOS-v0.80.8-Windows-Deployment.zip
```

Before starting:

- [ ] Unzip the Windows deployment package.
- [ ] Keep the package available on the Backoffice / Server PC and Register #1.
- [ ] Confirm Windows Administrator access on the Backoffice / Server PC.
- [ ] Confirm Windows Administrator access on Register #1.
- [ ] Record the Backoffice / Server PC name and LAN IP.
- [ ] Confirm Register #1, iPad, and Counterpoint PC can reach the same store network.
- [ ] Confirm Epson receipt printer IP or Windows printer name.
- [ ] Confirm scanner is in HID keyboard mode.
- [ ] Confirm payment terminal device codes and provider credentials if using Helcim.
- [ ] Keep the prior installer/build and a fresh database backup available for rollback.

Recommended:

- [ ] Put the Backoffice / Server PC on UPS power.
- [ ] Reserve static/DHCP IPs for the server PC and receipt printer.
- [ ] Record every generated password, station role, device IP, and install result in the deployment log.

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3. Backoffice / Server PC

Install the server first. Nothing else should be installed until this machine is healthy.

3.1 Install

1. On the Backoffice / Server PC, open the unzipped deployment package.
2. Run:

```
Start-RiversideDeployment.cmd
```

3. Choose:

```
Backoffice / Server
```

4. Click **Check**.
5. Resolve failed readiness checks.
6. Click **Install**.

The installer should:

- Install or use PostgreSQL.
- Create/update the Riverside database user.
- Create/update the Riverside database.
- Copy `riverside-server.exe`.
- Copy web/PWA files.
- Copy migrations and seed files.
- Apply migrations.
- Download the pinned ROSIE LLM model (Gemma E4B, ~5.4 GB) and install the sherpa-onnx Python voice stack.
- Write `C:\RiversideOS\server\.env`.
- Write server credential secrets into the Windows machine environment.
- Add the inbound firewall rule for port `3000`.
- Create the scheduled task `Riverside OS Server`.
- Start Riverside Server.
- Check `http://127.0.0.1:3000/api/staff/list-for-pos`.

3.2 Password And Secret Notes

- If PostgreSQL already exists, enter the existing `postgres` admin password.
- If the manager installs PostgreSQL and password fields are blank, it can generate passwords.
- Riverside database password is generated when blank or placeholder.
- Riverside app secret is generated when blank, too short, or placeholder.
- `RIVERSIDE_CREDENTIALS_KEY` is required before Settings can save encrypted integration credentials.
- Helcim, Shippo, QBO, Podium, Counterpoint, and other provider credentials should be entered in Backoffice Settings after the server is healthy.
- Do not use local `.env` edits as the normal way to configure integrations. Use the Windows repair tools only when the server credential key or Counterpoint bridge token needs recovery.
- Do not commit or publicly share `riverside-deployment.config.json`.

3.3 Server Acceptance

On the Backoffice / Server PC:

- [ ] Riverside opens locally.
- [ ] Staff sign-in screen loads.
- [ ] API host is `http://127.0.0.1:3000`.
- [ ] Settings → Updates shows Riverside `0.80.8`.
- [ ] Settings → General → About this build shows the expected server/API.
- [ ] Settings → Data & Backups opens.
- [ ] Backup path is writable.
- [ ] Server restart works through the Deployment Manager or the `Riverside OS Server` scheduled task.
- [ ] Deployment Manager PostgreSQL Status panel shows service **running**, connection **OK**, and the `riverside_os` database exists.

From another device on the store network:

- [ ] Open `http://SERVER-IP:3000`.
- [ ] Staff sign-in screen loads.
- [ ] No stale/update mismatch warning appears.

3.4 Server Repair Tools

Use these only when the symptom matches.

Tool	Use when
Deployment Manager **Repair**	Server files, firewall, scheduled task, migrations, or `.env` need to be refreshed without resetting the database.
Deployment Manager **PostgreSQL Status**	Check PG service state, connectivity, version, DB size, and migration count. Use **Start PG** / **Restart PG** / **Stop PG** buttons to control the service.
Deployment Manager **Stop Server**	Cleanly stop the Riverside OS Server task and process.
Deployment Manager **Uninstall Server**	Remove the server binary, task, and firewall rule while preserving the database.
`Install-RosieAiStack.cmd`	Installs the ROSIE AI models and Python voice tools, patches the server `.env`, and restarts the server without a full reinstall.
`Repair-RiversideCredentialsKey.cmd`	Settings says `RIVERSIDE_CREDENTIALS_KEY must be set` before integration credentials can be saved.
`Set-CounterpointBridgeToken.cmd`	Counterpoint bridge reaches Riverside but fails with `health 401`.
`Reset-RiversideDatabase.cmd`	Fresh pre-live install is badly interrupted and you intentionally need to reset the Riverside database. Do not use on live production data without rollback approval.

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4. Integration Credentials and External APIs

Do this after the server is installed and before relying on Helcim, QBO, Counterpoint, Podium, Meilisearch, or other integration credentials in production.

4.1 Credential Encryption Key Check

Riverside OS encrypts sensitive integration credentials in the database. A system encryption key must be established before settings can be saved.

- [ ] Sign in as an administrator on any workstation.
- [ ] Navigate to **Settings** and click on any integration (e.g., Helcim, QBO).
- [ ] Attempt to save a configuration. If a warning says `RIVERSIDE_CREDENTIALS_KEY must be set`:
  - [ ] Go to the Backoffice / Server PC.
  - [ ] Open the deployment package folder.

- [ ] Run `Repair-RiversideCredentialsKey.cmd` with administrator rights.
- [ ] Verify the script elevates, generates the key, adds it to the system environment variables, and restarts the Riverside OS Server service.
- [ ] Re-login to the app and confirm integrations can now be saved without warning.

4.2 QuickBooks Online (QBO) Setup

QuickBooks Online synchronizes sales invoices, returns, cash roundings, payments, and inventory receiving transactions to keep the store ledger in parity.

- [ ] In Back Office, navigate to **Settings → Integrations → QuickBooks**.
- [ ] Verify the **Client ID** and **Client Secret** are entered.
- [ ] Click **"Connect to QuickBooks"** to launch the secure OAuth2 authorization flow.
- [ ] Sign in with the store's primary QBO account and select the correct Company file.
- [ ] Authorize connection. Confirm that redirection back to the Riverside OS portal succeeds and displays a green `Connected` badge.
- [ ] Configure the accounts mappings:
  - [ ] Mapped **Sales Income Account** (e.g., Retail Sales).
  - [ ] Mapped **Cash/Tender Deposit Bank Account** (e.g., Undeposited Funds / Main Operating Bank).
  - [ ] Mapped **Sales Tax Liability Account** (e.g., Sales Tax Payable).
  - [ ] Mapped **Accounts Receivable (AR) Account** for layaway bookings.
  - [ ] Mapped **Inventory Asset Account** and **COGS Account**.
- [ ] Confirm **Transactional Outbox** state:
  - [ ] Verify the QBO Sync Worker status is `Idle` or `Syncing`.
  - [ ] Trigger a test sync on a single historical invoice; verify receipt in the QBO sandbox or ledger.

4.3 Helcim Card Payment Integration

Helcim is the integrated terminal provider for physical card swiping and capture at retail checkout.

- [ ] In Back Office, navigate to **Settings → Integrations → Helcim**.
- [ ] Enter the **Merchant ID** and **API Access Token**.
- [ ] Verify Environment is set to **Production** (unless performing pre-launch sandbox test runs).
- [ ] Configure terminal hardware matching:
  - [ ] Set up Register IP or terminal code for the physical Helcim Smart Terminal on the station.
  - [ ] Trigger a test ping from the register interface; verify response displays `Terminal Connected`.

- [ ] Webhook Verification:
  - [ ] Copy the webhook endpoint URL: `http://SERVER-IP:3000/api/webhooks/helcim`.
  - [ ] Log in to the Helcim Developer Portal and configure this URL under webhook receivers.
  - [ ] Subscribe to events: `order.approved`, `transaction.captured`.
  - [ ] Trigger a $0.01 test transaction to confirm callback resolves successfully.

4.4 Shippo Carrier Integration

Shippo is used to pull real-time shipping carrier rates, buy labels, and track fulfillment packages.

- [ ] In Back Office, navigate to **Settings → Integrations → Shippo**.
- [ ] Enter the **Shippo API Key** (must start with `shippo_prod_` for live launch).
- [ ] Enter the **Default Store Origin Address** (used as dispatch location for shipping quotes).
- [ ] Click **Save**.
- [ ] Go to **Fulfillment Orders → Shipments Hub** and verify that active carriers (USPS, UPS, FedEx) load and show rate quotes.

4.5 Podium Texting & Feedback Integration

Podium drives automated SMS customer communication, text-to-pay drawers, and reviews.

- [ ] In Back Office, navigate to **Settings → Integrations → Podium**.
- [ ] Enter the **Podium API Token** and **Location ID**.
- [ ] Verify webhook receiver status is `Active` and subscribed to incoming message events.
- [ ] Verify **Staff Identity Mapping**:
  - [ ] Map Riverside Staff Tracking IDs to the corresponding Podium User IDs in the mapping matrix.
  - [ ] Verify that texts sent through the CRM panel reflect the correct logged-in cashier.

4.6 Meilisearch Setup & Index Hydration

Meilisearch provides high-speed typotolerant local search for catalog products and customers.

- [ ] Verify the Meilisearch database service is running on Port `7700`.
- [ ] In Back Office, navigate to **Settings → Integrations → Meilisearch**.
- [ ] Confirm **Meilisearch URL** is configured (typically `http://127.0.0.1:7700`).
- [ ] Enter the **API Key** (Master/Search token).
- [ ] Click **"Test Connection"** and confirm status is `Connected`.
- [ ] Click **"Full Reindex"** to push all database products, customers, and active orders to Meilisearch.
- [ ] Go to the main search bar and verify queries return instantly.

4.7 Counterpoint Bridge Setup

NCR Counterpoint Node Bridge transfers legacy transactions, customer history, and catalog data.

- [ ] In Back Office, navigate to **Settings → Integrations → Counterpoint**.
- [ ] Copy the generated **Bridge sync token**.
- [ ] On the Counterpoint PC, open `C:\counterpoint-bridge\.env`.
- [ ] Set `ROS_BASE_URL=http://SERVER-IP:3000`.
- [ ] Set `COUNTERPOINT_SYNC_TOKEN` to match the token from Settings.
- [ ] Restart the Counterpoint Node Bridge service.
- [ ] In Back Office, verify the Counterpoint dashboard shows the status is `Connected`.
- [ ] Verify the mapping configurations:
  - [ ] Mapped Categories.
  - [ ] Mapped Payment Tenders (including Sverige cash rounding mapping).
  - [ ] Mapped Staff link profiles.
- [ ] Run a manual status query to confirm sync is listening without database warnings.

4.8 Metabase Analytics Setup

Metabase provides executive dashboards, sales charts, inventory metrics, and alterations tracking. It runs on Port `3001` and connects using the read-only PostgreSQL role `metabase_ro`.

- [ ] Verify the Metabase container is running (`docker ps` shows `riverside-os-metabase` is active on port `3001`).
- [ ] Connect the database in Metabase:
  - [ ] Open Metabase Admin → Databases.
  - [ ] Configure/verify the Riverside Postgres connection (host `db-primary` or local, database `riverside_os`, user `metabase_ro`).
  - [ ] Confirm Metabase is set to query the `reporting` schema as its primary data source.
- [ ] Run the metadata sync script:
  - [ ] On the Backoffice / Server PC, run the sync script:

```
node scripts/metabase-refresh-reporting-metadata.mjs
```

  - [ ] Confirm the script logs connection success, syncs schemas, and successfully models the core fields for `reporting.order_lines` and `reporting.payment_ledger`.
  - [ ] In Metabase Databases Admin, verify that a manual **Sync database schema** and **Re-scan field values** complete cleanly.

- [ ] Set up Collection governance:
  - [ ] Create collection **`Staff Approved`** (read-only for standard staff; contains no cost/margin fields).
  - [ ] Create collection **`Admin Financial`** (restricted to admin users; contains cost, margin, and payment reconciliation details).
- [ ] Create Metabase User Groups:
  - [ ] Create group **`Reporting - Staff`** (grant view access to `Staff Approved` only).
  - [ ] Create group **`Reporting - Admin`** (grant full access to both `Staff Approved` and `Admin Financial`).
- [ ] Verify core views are modeled and technical UUID columns are hidden (following `docs/METABASE_ADMIN_SETUP_STEPS.md`).
- [ ] In Back Office, navigate to **Insights** and verify the Metabase iframe loads successfully (validating the `metabase-launch` JWT SSO route).

4.9 ROSIE Local AI Copilot & Voice Stack

ROSIE handles AI Help Center Q&A, documentation searching, and operations analysis.

- [ ] Confirm the Gemma LLM model is downloaded and located in the model folder: `C:\RiversideOS\models\gemma-e4b.gguf`.
- [ ] Verify the server `.env` contains `RIVERSIDE_LLAMA_MODEL_PATH=C:\RiversideOS\models\gemma-e4b.gguf`.
- [ ] Verify Python voice synthesis stack is active (e.g. `sherpa-onnx` voice package).
- [ ] Open the Help Center search drawer and ask ROSIE a trial question. Confirm response is returned correctly.

4.10 Integration Troubleshooting Quick Reference

Integration	Diagnostic Check	Common Solution
QBO	Settings shows `Disconnected`	Re-auth via "Connect to QuickBooks". Check if credentials expired.
Helcim	Terminal times out during checkout	Verify terminal IP hasn't changed. Confirm terminal is on the same subnet.
Shippo	`Carrier quote error 401`	Shippo API key is invalid or has been revoked. Replace token.
Podium	Text messages fail to deliver	Check Location ID matching in settings. Verify account has available credits.
Meilisearch	Search returns zero matches	Confirm Meilisearch service is running. Click "Full Reindex" in Settings.
Counterpoint	Dashboard shows `health 401` or `503`	Tokens mismatch. Run `Set-CounterpointBridgeToken.cmd` on server.
Metabase	Insights iframe fails to load or permissions error	Verify Metabase container on port 3001. Check database connection for `metabase_ro` user. Ensure staff has the correct Metabase group permissions.
ROSIE	Help Copilot returns empty or fails	LLM file missing or wrong path. Check `RIVERSIDE_LLAMA_MODEL_PATH` in `.env`.

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5. Register #1 Windows PC

Install Register #1 after the Backoffice / Server PC passes acceptance.

5.1 Install

1. On Register #1, open the unzipped deployment package.
2. Run:

```
Start-RiversideDeployment.cmd
```

3. Choose:

```
Register #1
```

4. Enter the Backoffice / Server PC API base.

Correct examples:

```
http://SERVER-PC-NAME:3000
http://10.64.70.196:3000
```

Do not use:

```
http://127.0.0.1:3000
```

unless Riverside Server is running on Register #1 itself.

5. Enter Register #1 station label.
6. Choose receipt printer mode and cash drawer setting.
7. Click **Check**.
8. Resolve failed checks.
9. Click **Install**.

The installer should:

- Write `C:\ProgramData\RiversideOS\station-config.json`.
- Install the Riverside desktop app.
- Save the API base.
- Save printer and cash drawer settings.
- Launch the app when possible.

5.2 First Launch

- [ ] Open the Riverside desktop app.
- [ ] Confirm the app icon is the Riverside logo, not an old placeholder.
- [ ] Confirm Settings → Updates shows `0.80.8`.
- [ ] Confirm Settings → General → About this build shows `http://SERVER-IP:3000`.
- [ ] Staff sign-in works.
- [ ] Open Register #1.
- [ ] Enter the opening float according to store procedure.
- [ ] Confirm Register #1 is the cash drawer lane.

6. Receipt Printer, Cash Drawer, And Scanners

Do hardware validation on Register #1 before iPad/PWA validation.

6.1 Receipt Printer

Preferred receipt printer setup:

```
Mode: Network address
Printer: Epson TM-m30III
Port: 9100
IP: static or DHCP-reserved Epson IP
```

Use **Installed printer on this PC** only for USB/driver-managed fallback.

PWA receipt printing is still direct hardware printing. The iPad/PWA sends the request to Riverside Server, and Riverside Server opens the Epson printer IP/port. It should not open the browser print dialog for receipts.

Checklist:

- [ ] Epson receipt printer has paper.
- [ ] Epson printer IP or Windows printer name is recorded.
- [ ] Printers & Scanners settings are saved.
- [ ] Test receipt prints.
- [ ] PWA receipt test prints from the configured Epson IP/port.
- [ ] Register #1 receipt reprint path works.

6.2 Cash Drawer

- [ ] Cash drawer is connected to the receipt printer.
- [ ] Cash drawer opens for CASH/CHECK tender.
- [ ] Cash drawer does not intentionally open for card-only tender or receipt reprint.
- [ ] Staff understand physical cash belongs to Register #1.

6.3 Scanner

- [ ] Scanner is in HID keyboard mode.
- [ ] Scanner sends Enter/Tab suffix if desired.
- [ ] POS product search receives scanned text.
- [ ] Inventory/search input receives scanned text.

- [ ] Charging/spare scanner plan is known.

6.4 Reports And Tags

- [ ] Report printing uses browser/Windows print and the selected report printer.
- [ ] Clothing tags use the Zebra 2844 station by installed printer name or network IP.
- [ ] Zebra network checks fail clearly if Riverside Server cannot reach the tag printer.

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7. Register #2 iPad PWA

Register #2 is a satellite lane. It uses the Backoffice / Server PC URL while inside the store.

7.1 Prepare iPad

- [ ] iPad is on the same store Wi-Fi/network as the Backoffice / Server PC.
- [ ] iPad date/time is correct.
- [ ] Safari is available.
- [ ] Bluetooth scanner, if used, is paired as keyboard/HID input.

7.2 Install PWA

1. Open Safari.
2. Visit the Riverside server URL.

Examples:

```
http://SERVER-PC-NAME:3000
http://10.64.70.196:3000
```

3. Confirm the Riverside sign-in screen loads.
4. Tap **Share**.
5. Tap **Add to Home Screen**.
6. Name it:

```
Riverside Register 2
```

7. Launch Riverside from the new Home Screen icon.

Do not run Register #2 from a regular Safari tab for production use.

7.3 First Launch

- [ ] Launch from the Home Screen icon.
- [ ] Staff sign-in works.
- [ ] Settings → Updates shows `0.80.8`.
- [ ] No **Update incomplete** warning appears.
- [ ] Open/attach Register #2 only after Register #1 is open.
- [ ] Confirm it behaves as a satellite lane.
- [ ] Confirm customer lookup works.
- [ ] Confirm item lookup/cart works.
- [ ] Confirm checkout drawer fits comfortably on the iPad.
- [ ] Confirm staff know cash is handled at Register #1.

7.4 iPad Scanner And Receipt Path

- [ ] Pair Zebra CS6080 or similar as Bluetooth keyboard/HID.
- [ ] Open a product/SKU search field.
- [ ] Scan a barcode.
- [ ] Confirm scanned text appears in the focused field.
- [ ] If iPad receipt printing is expected, confirm the server can reach the network printer.
- [ ] If iPad print fails, reprint from Register #1.

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8. Other Back Office Devices

For Windows PCs that are not the server and not Register #1:

- [ ] Open the Riverside server URL in Edge/Chrome.
- [ ] Use browser install prompt if desired.
- [ ] Sign in with staff PIN.
- [ ] Confirm Customers loads.
- [ ] Confirm Orders loads.
- [ ] Confirm Inventory loads.
- [ ] Confirm Reports/Insights loads if the role has access.
- [ ] Confirm Settings → Updates shows `0.80.8`.
- [ ] Do not install the Backoffice / Server role on these PCs.

Use the Riverside desktop app only when native printer/scanner reliability is required.

9. Production Smoke Test

Run this before first customer use.

Server

- [ ] Reboot or restart the Backoffice / Server PC.
- [ ] Confirm `Riverside OS Server` starts automatically.
- [ ] Confirm sign-in works on the server PC.
- [ ] Confirm another device can open `http://SERVER-IP:3000`.
- [ ] Confirm backup path is writable.
- [ ] Deployment Manager PostgreSQL Status shows **running** and **OK**.

Register #1

- [ ] Staff sign-in works.
- [ ] Register #1 opens.
- [ ] Scanner works.
- [ ] Receipt prints.
- [ ] Cash drawer opens for CASH/CHECK.
- [ ] Card/Helcim flow is validated if enabled.
- [ ] Z-close path is understood.

Register #2 iPad

- [ ] Launches from Home Screen icon.
- [ ] Staff sign-in works.
- [ ] Register #2 attaches after Register #1 is open.
- [ ] Scanner works if used.
- [ ] Receipt fallback path is understood.
- [ ] No stale version warning.

Safe Sale Flow

Run one supervised low-risk flow:

- [ ] Sign in as staff.
- [ ] Open Register #1.
- [ ] Search or scan an item.

- [ ] Add item to cart.
- [ ] Open checkout drawer.
- [ ] Cancel safely or complete a store-approved test sale.
- [ ] Confirm receipt behavior.
- [ ] Confirm drawer behavior.
- [ ] Confirm the transaction appears in transaction history if completed.

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10. Updates And Hotfixes

After any update:

- [ ] Update Backoffice / Server PC first.
- [ ] Confirm server starts and Settings → Updates shows `0.80.8`.
- [ ] Update Register #1 desktop app.
- [ ] Relaunch Register #1 and confirm Settings → Updates shows `0.80.8`.
- [ ] On iPad, open Settings → Updates → iPad and browser app → Check app files.
- [ ] If the PWA update prompt appears, tap **Reload now** only when no sale is in progress.
- [ ] Confirm no station shows **Update incomplete**.

Never run sales if the server, Windows app, and PWA files show different Riverside versions.

Update package rule:

- Use the latest provided deployment zip for the specific version.
- Prefer the included `.cmd` helpers for targeted repairs.
- Do not reset the database as a troubleshooting shortcut.

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11. Backup And Rollback

Before live use:

- [ ] Create a fresh database backup.
- [ ] Confirm the backup file exists and has non-trivial size.
- [ ] Keep a copy off the server PC if practical.
- [ ] Keep the previous installer/build accessible.
- [ ] Record backup timestamp and file name.

- [ ] Record who has rollback authority.

Rollback rule:

1. Stop all registers.
2. Stop active checkouts.
3. Record the reason for rollback.
4. Restore the database backup before running older binaries against migrated data.
5. Do not run an older server against a newer migrated database without restore approval.

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12. Quick Troubleshooting

PWA will not load

1. Confirm the device is on the correct network.
2. Open `http://SERVER-IP:3000` in the device browser.
3. Check DNS/server name.
4. Check firewall on the Backoffice / Server PC.
5. If using HTTPS, check certificate/date/time.
6. Close and reopen the Home Screen icon.
7. If still stale, remove the icon, clear site data, and Add to Home Screen again.

Register app will not reach server

1. Check Settings → General → About this build for API base.
2. Confirm Register #1 is not pointed at `127.0.0.1` .
3. Open the Backoffice / Server PC URL from Register #1.
4. Run Repair from Deployment Manager if station config is wrong.

PostgreSQL is down or unreachable

1. Open the Deployment Manager.
2. Check the **PostgreSQL Status** panel.
3. If service shows **stopped**, click **Start PG**.
4. If connection shows **Failed**, verify PostgreSQL is installed and the service is running.
5. If `psql.exe` is not found, install PostgreSQL or set `psqlPath` in the config.
6. Use **Restart PG** if the service is running but connections are failing.

Receipt will not print

1. Check Printers & Scanners.
2. For network mode, confirm Epson IP and port `9100`.
3. For installed-printer mode, confirm Windows test page works.
4. Restart the desktop app after printer/network changes.
5. For iPad, confirm the server can reach the network printer.

Counterpoint bridge fails

1. Confirm `ROS_BASE_URL=http://SERVER-IP:3000`.
2. If bridge shows `health 503`, configure the token in Riverside or run `Repair-RiversideCredentialsKey.cmd`.
3. If bridge shows `health 401`, run `Set-CounterpointBridgeToken.cmd` on the server PC and paste the exact token from the bridge `.env`.
4. Restart the bridge after changing `.env`.

Update incomplete

1. Finish updating the Backoffice / Server PC.
2. Finish updating Register #1 desktop app.
3. On iPad/PWA, Check app files and reload.
4. Do not continue production work until all surfaces show `0.80.8`.

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13. DevOps Center (In-App)

For staff with `ops.dev_center.view` permission.

Access

1. Open Riverside on any device.
2. Sign in as a staff member with DevOps permissions.
3. Open **Settings**.
4. Navigate to **ROS Dev Center**.

What It Shows

- **Operational status** — DB health, integrations, stations online/offline
- **Runtime diagnostics** — memory, disk, service status
- **E2E health** — latest CI test lane status

- **Alerts** — open system alerts, ack/resolve actions
- **Stations** — register heartbeat status and versions
- **Bug reports** — pending bugs with server log snapshots
- **GitHub** — recent workflow runs, releases, one-click release builds

Trigger a Release Build (Manager Access Required)

Requires `ops.dev_center.actions` permission and `RIVERSIDE_GITHUB_TOKEN` configured on the server.

1. In ROS Dev Center, scroll to **GitHub DevOps**.
2. Click **Build Release**.
3. Confirm the workflow dispatch.
4. Check `github.com/cpg716/riverside-os/actions` for the running workflow.

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14. Automated Dependency Maintenance

Riverside OS uses GitHub Actions for CI/CD. Do not disable these workflows.

Workflow	Purpose
`lint.yml`	`cargo fmt`, `cargo clippy`, `cargo audit` on every PR
`security-audit.yml`	Weekly `cargo audit` (CVE scan) + `cargo outdated` (stale deps)
`tauri-register-updater-release.yml`	Builds signed Windows installer on `v*` tag push
`dependabot-auto-merge.yml`	Auto-merges patch-level dependency updates

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15. Standalone ROS Dev Center (macOS)

A dedicated macOS companion app for development and remote management.

Build (on a Mac with Rust/Xcode)

```
cd ros-dev
npm install
```

```
npm run tauri build
```

The `.dmg` appears in `ros-dev/src-tauri/target/release/bundle/dmg/`.

Features

- **Auto-discovery** — scans Tailscale peers and local subnet for ROS servers via high-performance concurrent Rust TCP sweeps (avoiding CORS and browser connection limit limitations)
- **Keychain security** — stores staff Access PINs securely in the native system Keychain instead of plaintext `localStorage`
- **IP route shielding** — restricts access to administrative `/api/ops/*` routes to loopback, RFC 1918 private subnets, and Tailscale IPs
- **Server profiles** — save and switch between dev/staging/production instantly (stripping secrets from disk serialization)
- **Tailscale detection** — shows connection status and tailnet name
- **Real-time dashboard** — DB health, stations, alerts, bugs, workflows, releases
- **GitHub integration** — view workflow runs, releases, trigger builds
- **ROSIE AI analysis** — one-click diagnostic analysis via local Gemma LLM
- **External AI support** — copy diagnostic prompts for ChatGPT/Claude/Cursor

Connect

1. Launch the app.
2. Check Tailscale status badge.
3. Click **"Scan for Riverside Servers"** to auto-discover.
4. Click a discovered server or select a saved profile.
5. Enter your staff PIN.
6. Dashboard auto-refreshes every 30 seconds.

Diagnostics & AI

1. Click **"Run Diagnostics"** in the dashboard.
2. Review server version, DB pool, errors, warnings.
3. Click **"Analyze with ROSIE"** for AI-powered fix suggestions.
4. Or click **"Copy"** to use external AI tools.

Use Cases

- Remote DevOps monitoring via Tailscale from anywhere
- Trigger release builds without browser sign-in

- View station health, alerts, and workflow status
- AI-assisted debugging with file-level fix recommendations

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16. Deployment Log

Item	Value
Deployment date	
Release version	`0.80.8`
Backoffice / Server PC name	
Backoffice / Server PC IP	
Server API host on server PC	`http://127.0.0.1:3000`
Register #1 PC name	
Register #1 API base	
iPad Register #2 URL	
Counterpoint PC name	
Counterpoint `ROS_BASE_URL`	
Epson receipt printer IP/name	
Zebra scanner model(s)	
Backup file name	
Backup timestamp	
Installer owner	
Store manager signoff	
Rollback authority	